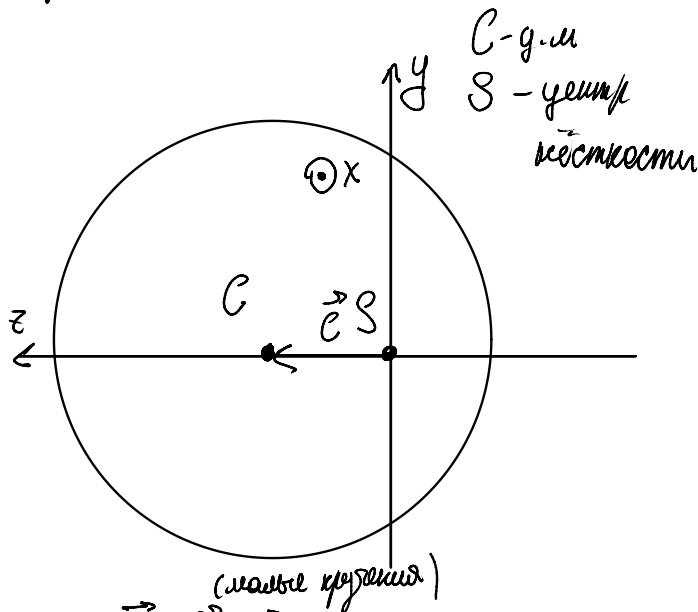


1) Для сегмента с крив. симметрией  
центр жесткости и центр масс не  
совпадают



$$\vec{r}_e \approx \vec{r}_s + \vec{e} + \varphi[\vec{x}, \vec{e}]$$

$$\vec{r}_s = (0, uH, 0)$$

$$[\hat{x}, \vec{c}] = (0, -c_z, c_y)$$

$$\vec{r}_c = (0, 4 - \rho e_z, \frac{\rho}{g})^0$$

$$y_e = U - \varphi C_z \Rightarrow \dot{y}_e = \dot{U} - C \dot{\varphi}$$

$$dT_{\text{сеч}} = \frac{1}{2} (dm \dot{y}_c^2 + dI_c \dot{\varphi}^2)$$

$$dm = \rho S dx, dI_c = \rho I_p dx$$

$$dT_{\text{сеч}} = \frac{1}{2} dm \dot{u}^2 - dm c \dot{u} \dot{\varphi} + \frac{1}{2} (dI_c + dm c^2) \dot{\varphi}^2$$

$$T_{\text{сеч}} = \frac{1}{2} \int_0^L \rho S \dot{u}^2 - \rho S c \dot{u} \dot{\varphi} + \rho I_p \dot{\varphi}^2 dx$$

2) Поиск ном.-энергии

$$u_x(y, z, t, x)$$

$$u_y(y, z, t, x)$$

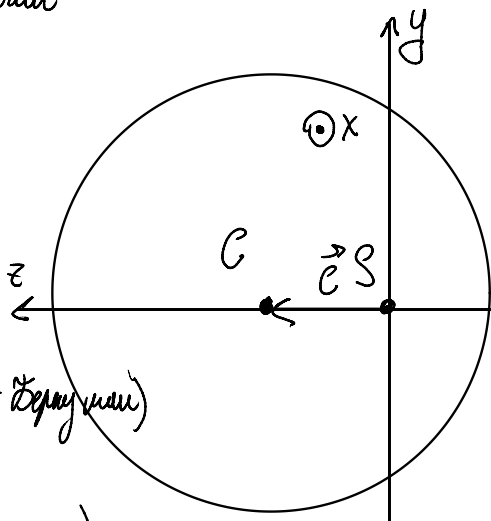
$$u_z(y, z, t, x)$$

$$u_y \propto w(x, t) \quad (\text{Зависит от } x \text{ и } t)$$

$$u_x = -y \partial_x w$$

(Вектор  $S$  нулевой)

$$u_y = w(x, t) - z \varphi(x, t), u_z = y \varphi(x, t)$$



$$\varepsilon_{xx} = \partial_x u_x = -y \partial_x^2 w$$

$$\partial_y u_y = 0 = \partial_z u_z$$

Кривизны даём сбаланс

$$\chi_{x\theta} = r \frac{\partial \varphi}{\partial x} \quad r - \text{расст. по } OX, \theta - \text{угл. коэф.}$$

$$\chi_{xr} = -z \frac{\partial \varphi}{\partial x}$$

$$\chi_{kz} = y \frac{\partial \varphi}{\partial x}$$

$$\varepsilon_{yy} = \varepsilon_{zz} = -\nu \varepsilon_{xx} = -\nu y \partial_x^2 w$$

Законы Гука

$$\sigma_{xx} = E \varepsilon_{xx} = -E y \partial_x^2 w$$

$$\tau_{xy} = G \chi_{xy} = -G z \partial_x \varphi$$

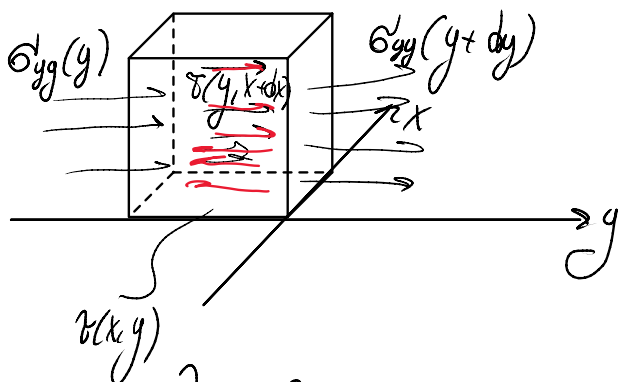
$$\tau_{xz} = G y \partial_x \varphi$$

Из условий дол. равн-ия

$$\sigma'_{yy} = \sigma'_{zz} = 0. \text{ Это верно для}$$

кв. стержней, но я считаю это верным и  
 дам кручение стержней в силу его тонкости

Рассм. элем. стержня



$$\partial_x \delta + \partial_y \sigma_{xy} = 0$$

$$\sigma_{xy} \sim \delta_{xy} \frac{h}{L} - \text{мало} \Rightarrow$$

$$\Rightarrow \sigma_{xy} \times \sigma_{xz} = 0$$

$$U = \int u dV$$

$$u = \frac{1}{2} (\sigma_{xx} \epsilon_{xx} + \sigma_{yy} \epsilon_{yy} + \sigma_{zz} \epsilon_{zz} + \tau_{xy} \gamma_{xy} + \tau_{xz} \gamma_{xz}) = \frac{1}{2} E y^2 (\partial_x^2 w)^2 + \frac{1}{2} G (x^2 + y^2) (\partial_x \varphi)^2$$

$$u = \frac{1}{2} E y^2 (\partial_x^2 w)^2 + \frac{1}{2} G r^2 (\partial_x \varphi)^2$$

A-mul. over.

$$U = \frac{1}{2} \left( \int_0^L E (\partial_x^2 w)^2 \int_A y^2 dA + \right.$$

$$\left. + \int G (\partial_x \varphi)^2 \int r^2 dA \right)$$

$$I_z = \int_A y^2 dA \quad J = \int_A r^2 dA$$

$$U = \frac{1}{2} \left( \int_0^L E I_z (\partial_x^2 w)^2 dx + \int_0^L G J (\partial_x \varphi)^2 dx \right)$$

$$T = \frac{1}{2} \int_0^L [\rho S \dot{w}^2 - \rho g S e \dot{w} \dot{\varphi} + \rho \bar{I}_p \dot{\varphi}^2] dx$$

$$U = \frac{1}{2} \int_0^L [E \bar{I}_z (\partial_x^2 w)^2 + G J (\partial_x \varphi)^2] dx$$

$$\mathcal{L} = \frac{1}{2} \rho S (\dot{w})^2 - \rho g S e \dot{w} \dot{\varphi} + \frac{1}{2} E \bar{I}_z (\partial_x^2 w)^2 - \frac{1}{2} G J (\varphi')^2$$

Après le Lagrangien quel nous

1)  $w(x, t)$

$$\frac{\partial \mathcal{L}}{\partial w} - \partial_t \left( \frac{\partial \mathcal{L}}{\partial \dot{w}} \right) - \partial_x \left( \frac{\partial \mathcal{L}}{\partial \partial_x w} \right) + \partial_x^2 \frac{\partial \mathcal{L}}{\partial \partial_x^2 w} = 0$$

2)  $\varphi(x, t)$

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_t \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} - \partial_x \frac{\partial \mathcal{L}}{\partial \partial_x \varphi} = 0$$

$$\frac{\partial \mathcal{L}}{\partial w} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{w}} = \rho \delta \ddot{w} - \rho \delta \epsilon \dot{\varphi}$$

$$\partial_t \frac{\partial \mathcal{L}}{\partial \dot{w}} = \rho \delta \dot{w} - \rho \delta \epsilon \dot{\varphi}$$

$$\frac{\partial \mathcal{L}}{\partial \partial_x w} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \partial_x^2 w} = -EI_z \partial_x^2 w$$

$$\partial_x^2 \frac{\partial \mathcal{L}}{\partial (\partial_x^2 w)} = -EI_z \partial_x^4 w$$

$$\rho \delta \ddot{w} - \rho \delta \epsilon \dot{\varphi} + EI_z \partial_x^4 w = 0$$

$$\rho I \delta \ddot{\varphi} - \rho \delta \epsilon \ddot{w} - GJ \partial_x^2 \varphi = 0$$

Мы делаем стандартный шаг

$$w = W(x) e^{i\omega t} \quad \varphi = \Phi(x) e^{i\omega t}$$

$$-\rho \delta \omega^2 W + \rho \delta \varepsilon \omega^2 \Phi + EI_z \partial_x^4 W = 0 \quad (1)$$

$$-\rho I_p \omega^2 \Phi + \rho \delta \varepsilon \omega^2 W - GJ \partial_x^2 \Phi = 0 \quad (2)$$

Гр.-ум:

$$\begin{aligned} w(0) = w(L) &= 0 \\ w'(0) = w'(L) &= 0 \\ \varphi(0) = \varphi(L) &= 0 \end{aligned}$$

Разложение по собств. функциям

$$(\Phi = 0) \quad EI_z \partial_x^4 U_n = \rho \delta \omega_{n,n}^2 U_n \quad (3)$$

Собств. функции, частоты  $U_n(x)$   $\omega_{n,n}$   
 Изгибание ( $W = 0$ )

$$-GJ \Phi_m'' = \rho I_p \omega_{\varphi,m}^2 \Phi_m$$

Вблизи резонанса полагаем

$$\omega_{n,n} \approx \omega_{\varphi,m} \equiv \omega$$

$$\begin{aligned} W(x) &= A U_n(x) \\ \Phi(x) &= B \Phi_m(x) \end{aligned} \quad \left( \text{при } \delta \text{ малом} \right)$$

Объяснение:

$$W(x) = A_1 U_1(x) + A_2 U_2(x) + \dots$$

Разложение по базису есобств.  
решений. В резонансе есть прецессия

Гамильтона (1) на  $U_n$ , интегрирую по  $dx$

$$- \rho S \omega^2 A \underbrace{\int U_n^2 dx}_{1(\text{нормировка})} + \rho S e \omega^2 B \underbrace{\int U_n \Phi_m dx}_{= \langle U_n, \Phi_m \rangle} +$$

$$+ EI_z A \int \underbrace{U_n \partial_x^4 U_n dx}_{=} = 0$$

$$\text{Из (3)} \quad EI_z \lambda_n^4 = \rho S \omega_{u,n}^2$$

$$(\omega_{u,n}^2 - \omega^2) A + e \omega^2 \langle U_n, \Phi_m \rangle B = 0$$

Аналогично для (2), учитывая  $\rho I_p$

$$(\omega_{\Phi,m}^2 - \omega^2) B + \frac{S}{I_p} e \omega^2 \langle \Phi_m, U_n \rangle A = 0$$

$$\gamma_1 = e \langle U_n, \Phi_m \rangle$$

$$\gamma_2 = \frac{S}{I_p} e \langle \Phi_m, U_n \rangle$$

$$\omega_u = \omega_{u,n} \quad \omega_\Phi = \omega_{\Phi,m}$$

$$(\omega_u^2 - \omega^2)A + j_1 \omega^2 B = 0$$

$$(\omega_\varphi - \omega^2)B + j_2 \omega^2 A = 0$$

$$\vec{z} = \begin{pmatrix} \omega_u^2 - \omega^2 & j_1 \omega^2 \\ j_2 \omega^2 & \omega_\varphi^2 - \omega^2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = 0$$

Система имеет нетривиальное решение  $\exists \Leftrightarrow \det \vec{z} = 0$ ;  $j = j_1 j_2$

$$(\omega_u^2 - \omega^2)(\omega_\varphi^2 - \omega^2) - j^2 \omega^4 = 0$$

$$(1 - j^2)\omega^4 - (\omega_u^2 + \omega_\varphi^2)\omega^2 + \omega_u^2 \omega_\varphi^2 = 0$$

1)  $j = 0$ :  $\omega^2 = \omega_u^2$  или  $\omega^2 = \omega_\varphi^2$

Вдвух резонансах  $\omega_u \approx \omega_\varphi \approx \omega_0$  — два корня одинаковые.

2)  $j \neq 0$ ,  $\omega_{1,2}^2 = \omega_0^2 = \omega_\varphi^2$

$$(1 - j^2)\omega^4 - 2\omega_0^2 \omega^2 + \omega_0^4 = 0$$

$$\omega_1^2 = \frac{\omega_0^2}{1 - j} \quad \omega_2^2 = \frac{\omega_0^2}{1 + j}$$

$$\omega_{1,2} = \omega_0 (1 \pm j)$$

$$f_{1,2} = f_0 \left( 1 \pm \frac{f}{f_0} \right)$$

$$f_1 = 4070 \text{ PHz}$$

$$f_2 = 4175 \text{ PHz}$$

$$f_0 = \frac{\omega_1 + \omega_2}{2} = 4123 \text{ PHz}$$

$$f_{f_0} = \frac{\omega_2 - \omega_1}{2} = 53 \text{ PHz} \quad \Bigg| \Rightarrow$$

$$\Rightarrow \gamma = 1.3 \cdot 10^{-2}, \quad \frac{f}{f_0} \approx 6.4 \cdot 10^{-3}$$

$$(\sigma_f)^2 = (6.4 \cdot 10^{-3} f)^2$$